

Esthetic Considerations in Diagnosis and Treatment Planning

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Because beauty is primarily a matter of personal taste modified by social norms, visualizing beauty is a subjective experience. Creating a beautiful smile requires a foray into these subjective waters. This chapter provides comprehensive and, when possible, evidence-based guidelines that will enable dentists to provide esthetic as well as functional dentistry.

Esthetic Parameters

Facial height

The face can be divided vertically into thirds, and the length of the middle third of the face should approximately equal the lower third of the face when measured in repose[1] (Fig 3-1). The midface is measured from *glabella*, the most prominent point of the forehead between the eyebrows, to *subnasale*, the point below the base of the nose. The lower face is measured from subnasale to *soft tissue menton*, the lower border of the chin.

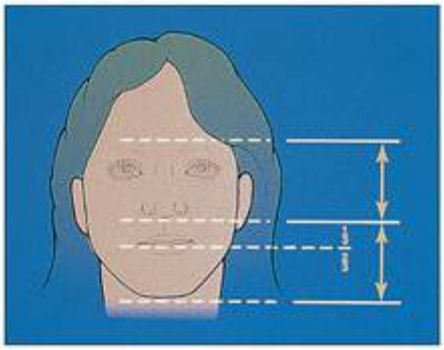


Fig 3-1 The length of the middle third of the face should equal the length of the lower third of the face.

Variations from the norm can reflect a continuum from underdevelopment to hypertrophic development of either one or both arches. However, regarding esthetic diagnoses that impact dental treatment, excessive length of the lower third of the face is most common. The long lower face is commonly the result of vertical maxillary excess and, in many cases, is accompanied by excess gingival display in the maxilla during full smile.

Lip length

The length of the upper lip is measured from subnasale to the inferior border of the upper lip in repose (Fig 3-2a). The average length of the upper lip is 20 to 22 mm in the young adult female and 22 to 24 mm in the young adult male.[2] The upper lip tends to lengthen with age. After age 40 years, the amount of incisal edge display decreases approximately 1 mm per decade.[3,4] When a patient presents with excess gingival display (more than 2 mm of gingiva exposed above the maxillary central incisors during full smile), lip length may be part of the etiology (Fig 3-2b).

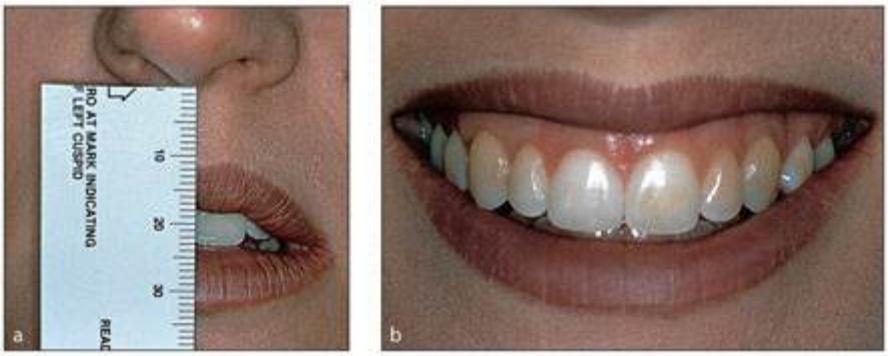


Fig 3-2*(a)* Lip length of 18 mm (average, 20 to 22 mm in women) causes gingival display in full smile. *(b)* Approximately 2 mm of gingival display in full smile.

Lip mobility

Mobility of the upper lip is measured from repose position to high smile position. The average lip mobility is 6 to 8 mm. In the patient with excess gingival display in full smile, hypermobility of the upper lip may be a contributing factor.

Symmetry

Outline symmetry is essential at the midline5; the maxillary central incisors should be mirror images of each other. Additionally, a line drawn between the maxillary central incisors should be perpendicular to the horizon[6] (Fig 3-3). Ideally, the maxillary dental midline should be coincident with the facial midline.[7] However, a 1- to 2-mm midline discrepancy has been shown to have a minimal effect on the overall beauty of the smile.[8] Asymmetry at the midline creates a visual tension in the observer, resulting in an unacceptable esthetic presentation. As the eye moves peripherally from the midline, deviations from perfect symmetry (eg, notched edges or slight differences in edge lengths) become acceptable, even desirable.[9]



Fig 3-3 Midline canted in relation to the horizon.

Incisal plane

The incisal plane should be parallel to the horizon; the interpupillary line is helpful in making this determination,[9] but only if the interpupillary line is parallel to the horizon. The incisal plane is evaluated from cusp tip of the maxillary right canine to cusp tip of the maxillary left canine. Although the incisal plane must be parallel to the horizon, it is generally not flat but instead has a curve that parallels the curve of the lower lip in full smile. In addition, it should not be canted up or down from right to left.

It is important not to perpetuate or create a canted incisal plane with restorations. If the interpupillary line is parallel to the horizon, the corners of the mouth should be pulled outward so that the upper lip parallels the interpupillary line (Fig 3-4). The relationship between the incisal plane and the interpupillary line, via the upper lip,

can then be visualized. To transfer this information to an articulator, a facebow may be used as long as the horizontal member of the facebow is made parallel to the horizon before attaching the bite fork. An incisal plane relationship bite may also be used. A bite registration paste is placed between the maxillary and mandibular incisors. A long cotton-tipped applicator is then embedded in the bite registration paste and set parallel to the horizon (Fig 3-5). This relationship bite can then be used to mount the maxillary cast with an accurate incisal plane orientation to the maxillary member of the articulator (Fig 3-6).



Fig 3-4 The upper lip is pulled to parallel the inter-pupillary line. The vermilion border of the upper lip is then used to evaluate the cant of the incisal plane and posterior occlusal plane in relation to the horizon.



Fig 3-5 Bite registration paste is placed between the maxillary and mandibular anterior teeth, and a cotton-tipped applicator is embedded in the paste. The cotton-tipped applicator is then paralleled with the interpupillary line, and the paste is allowed to set.



Fig 3-6 The bite registration is placed on the maxillary cast, and the stick is set parallel to the maxillary member of the articulator to orient the cast for mounting.

Posterior occlusal plane

The buccal cusp tips of the maxillary posterior teeth should provide a visual progression from the canine cusp tips, with no step up or step down. In addition, the posterior occlusal plane should not be canted up or down from right to left.[6] It is important not to perpetuate or create a canted posterior occlusal plane. The same techniques used to ensure an accurate mounting of the incisal plane are used for the posterior occlusal plane.

Buccal corridor

The *buccal corridor* is the space between the buccal surfaces of the maxillary posterior teeth and the cheek. In full smile, the buccal corridor is almost filled with teeth. However, a minimal negative space frames the maxillary posterior teeth and is desirable.[6] Excess buccal corridor space (Fig 3-7) is usually due to a developmental problem, such as a constricted maxillary arch. Inadequate space in the buccal corridor is usually due to bulky posterior restorations.



Fig 3-7 A patient with excess space in the buccal corridors.

Lower lip

In full smile, the incisal edges of the maxillary anterior teeth ideally are cradled by the lower lip[10,11] (Fig 3-8). The smile may also be very pleasing when a space exists between the lower lip and the maxillary incisal edges, as long as the space is uniform from right to left (Fig 3-9). Conversely, none of the incisal edges of the maxillary anterior teeth should be concealed by the lower lip in full smile.

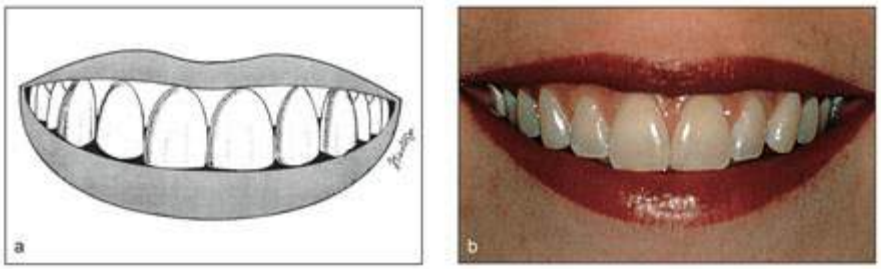


Fig 3-8*(a and b)* Ideal maxillary incisal edge position in relation to the lower lip and ideal relationship between the upper lip and the gingival line.

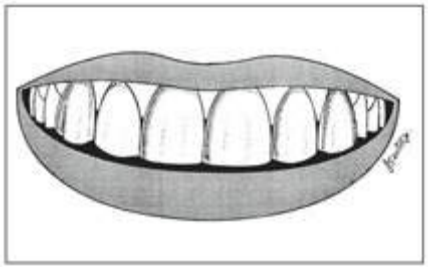


Fig 3-9 Uniform space between the maxillary incisal edges and the lower lip.

If there is a reverse incisal edge curve in relation to the lower lip or a significant space between the lower lip and the maxillary incisal edges during full smile, esthetics may be enhanced with increased incisal edge length. Conversely, if incisal edges of maxillary anterior teeth are hidden by the lower lip during full smile, there is likely a problem with the vertical position of the maxilla. The cause may be overeruption of maxillary anterior teeth, vertical maxillary excess, or both.

Upper lip

In full smile, the upper lip should ideally translate up to the gingival line[10] (see Fig 3-8b). This occurs in approximately 70% of the population. Approximately 10% have a high smile line, and approximately 20% have a low smile line.[11] To evaluate the gingival line, a straight line is drawn from the tooth-gingiva interface of the right maxillary canine to the tooth-gingiva interface of the left maxillary canine. The tooth-

gingiva interface of both central incisors should be on this line. The tooth-gingiva interface of the lateral incisors may either fall on the gingival line or be up to 1.5 mm below it[12] (Fig 3-10).

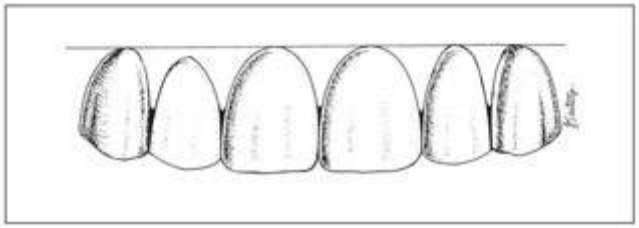


Fig 3-10 Gingival line from canine to canine. The maxillary lateral incisors can fall on this line or be up to 1.5 mm below it.

If the upper lip does not translate up to the gingival line during full smile, some of the clinical crowns of the maxillary anterior teeth remain covered. This results in a loss of dynamics in the smile. If, in full smile, the upper lip translates above the gingival line, the result is gingival display above the clinical crowns. Gingival display of 2 mm or more above the gingival line results in compromised esthetics.

Lip asymmetry

If a patient displays an upper lip asymmetry during full smile (Fig 3-11), it does not generally influence treatment. However, if a patient has an asymmetric upper lip translation, resulting in excess gingival display on one side, esthetic crown lengthening surgery may be accomplished to provide more symmetry in the posterior gingival display. Any time a patient has an upper lip asymmetry, he or she should be advised of the condition before restoration of the maxillary anterior teeth, because the brighter restored teeth will draw attention to the smile and accentuate the asymmetry.



Fig 3-11 Upper lip asymmetry.

If a patient has a lower lip asymmetry during full smile that results in a unilateral increase in negative space between the maxillary incisal edges and the lower lip, smile symmetry is lost (see Fig 3-31b). The treatment for the asymmetric lower lip

is Botox (Allergan).

Lip asymmetries, which can play a significant role in the final restorative result, are commonly overlooked.

Incisal edge placement of maxillary central incisors

Determining the correct position of the incisal edges of the maxillary central incisors is essential and the first step in the provision of anterior restorative dentistry. The following five guidelines are used to determine the correct incisal edge position:

1. In full smile, the incisal edges of the maxillary anterior teeth should be cradled by the lower lip[10,11] (see Fig 3-8b).
2. In full smile, the buccal cusp tips of the posterior maxillary teeth should provide a visual progression from the canine cusp tip, with no step up or step down[6] (Figs 3-12 and 3-13).

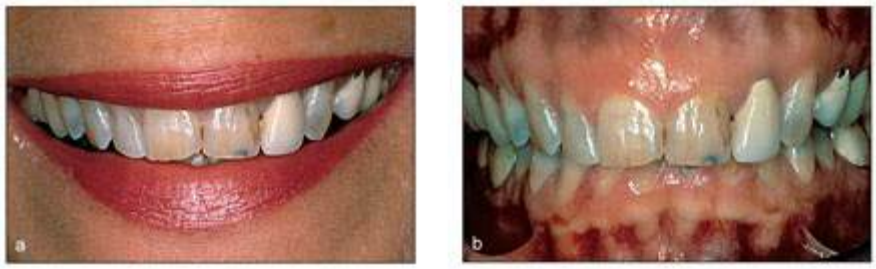


Fig 3-12*(a and b)* Note step up from maxillary left canine to maxillary left first premolar.



Fig 3-13 Note bilateral step down from maxillary canines to maxillary first premolars.

3. In gentle repose (have the patient say “M” or “Emma” and ask them to let their lips lightly fall apart), approximately 3 to 4 mm of the incisal edges of the maxillary central incisors should be displayed in the young adult female (Fig 3-14). In the young adult male, approximately 1 to 2 mm of the incisal edges should be displayed. After age 40 years, the amount of incisal edge display decreases approximately 1 mm per decade.[3]



Fig 3-14 Three millimeters of display of maxillary central incisors in repose.

4. The average length of the maxillary central incisor is 10 to 11 mm.[13]
5. When the patient says "F" or "V," the incisal edges of the maxillary central incisors should lightly touch the wet-dry border of the lower lip[14] (Fig 3-15).



Fig 3-15 Patient pronouncing an "F."

Steps 1 through 4 are used together to develop an approximation of the correct incisal edge position for the diagnostic wax-up. At this point, incisal edge position is dictated strictly by esthetics. After tooth preparation, provisional restorations that have been fabricated using the diagnostic wax-up are placed. The definitive incisal edge position is then developed dynamically, over time, in the provisional restorations to ensure suitable function and phonetics as well as esthetics. Step 5 is helpful in assessing phonetics with lengthened provisional restorations.

Facial contour of maxillary incisors

Divide the facial surface of the maxillary central incisor into two planes. The gingival half of the tooth should be parallel to and continuous in contour with the surface of the gingival tissue overlying the alveolus[15] (Fig 3-16). The incisal half is tapered back for ease in speaking and swallowing (Fig 3-17).

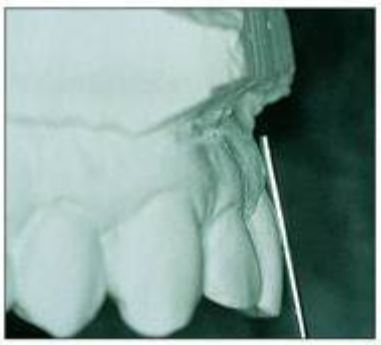


Fig 3-16 The gingival half of the maxillary central incisor is parallel to and continuous in contour with the surface of the gingival tissue overlying the alveolus.



Fig 3-17 The incisal half of the central incisor is contoured to feel comfortable to the patient during lip closure and speech.

Facial overcontouring of a partial- or full-coverage restoration in the gingival half can result in chronic gingival inflammation. Facial overcontouring in the incisal half may result in lip pressure, causing linguoversion of the overcontoured teeth or interference with the path of lip closure.

Lingual contour of maxillary incisors

Incorrect spacing between maxillary and mandibular anterior teeth may cause a lisp. A lisp can occur with too much or too little space, although it occurs most commonly with too little space.

If a patient develops a lisp after placement of provisional or definitive restorations, the position of the incisal edges of the mandibular incisors in relation to the maxillary central incisors when the patient makes an “S” sound must be determined. If the mandibular incisor approximates the cingulum or lingual concavity of the maxillary incisor, the lisp is corrected most commonly by increasing the lingual concavity of the maxillary incisors. If, however, the mandibular incisor approximates the incisal edge of the maxillary central incisor during the “S” sound,

the lisp can be corrected most commonly by changing the length of the maxillary central incisors.

Gingival zenith

The long axes of the maxillary anterior teeth are distally inclined. Therefore, the gingival contour adjacent to the maxillary incisors is not a symmetric, rounded arch form. Rather, the marginal gingiva has a parabolic shape with the highest point (*gingival zenith*) slightly distal to the midline of the tooth[12] (Fig 3-18).



Fig 3-18 The gingival zenith is slightly distal to the midline on the maxillary central and lateral incisors.

In gingival recontouring surgery, the gingival zenith should not be overemphasized. Although a distalized zenith is more common, many patients prefer a more symmetric gingival architecture.

Interproximal contact areas

Maxillary interproximal contact areas become progressively more gingival from central incisor to canine (Fig 3-19). The interproximal contact between the maxillary central incisors is in the incisal third of the teeth. However, the interproximal contact between the central and lateral incisors is at the junction of the incisal and middle thirds; it is slightly more gingival between the lateral incisors and the canines.[9]



Fig 3-19 Maxillary interproximal contact areas become progressively more gingival from central incisor to canine, and incisal embrasures increase in depth from midline to canine.

If the interproximal contact extends too far incisally, a closed and unnatural-appearing incisal embrasure results. If the interproximal contact does not extend far enough gingivally, an open gingival embrasure, or *black triangle*, results.

Incisal embrasures

The incisal embrasures increase from maxillary central incisor to canine (see Fig 3-19). While the incisal embrasure between the maxillary central incisors is minimal, the incisal embrasure between the maxillary central and lateral incisors is more pronounced and between the lateral incisors and canines is the most pronounced. Uniform incisal embrasures from maxillary canine to canine are esthetically unnatural.

Maxillary incisal edges and tooth morphology

In nature, it is impossible to determine sex based on tooth shape or incisal edge relationships.[11] However, tooth morphology and tooth-to-tooth relationships do convey information, albeit subjective, about the individual. In 1973, based on the writings of Frush and Fisher,[16,17] Lombardi[6] described relationships for fabricating complete dentures. His matrix is equally relevant for the dentulous patient today (Fig 3-20) and makes it possible to characterize the teeth in the diagnostic wax-up, the provisional restorations, and, ultimately, the definitive restorations.

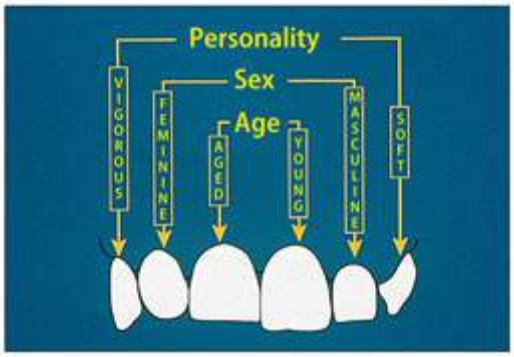


Fig 3-20 The Lombardi matrix describes characteristics associated with different incisal edge configurations.

Maxillary incisal edge shape (buccolingual)

Natural maxillary incisal edges, in a buccolingual direction, are not rounded but rather are sharp. Because of wear, the incisofacial line angle in adults is relatively sharp and blends into a 1-mm lingual facet before dropping off to the concave lingual surface (Fig 3-21). Rounded maxillary incisal edges give the restoration an unnatural appearance due to the light reflection off a curved surface.

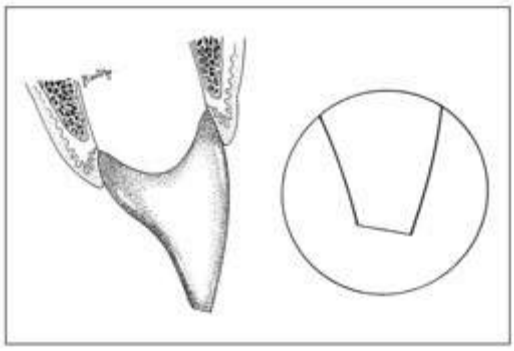


Fig 3-21 Maxillary incisal edge.

Halo effect

The natural incisal edge anatomy of the maxillary incisor commonly imparts a thin, white, opaque "halo effect" at the incisal edge that frames the incisal translucency (Fig 3-22). The halo effect is incorporated into porcelain restorations by building the sharp incisal edge anatomy into the crown or porcelain veneer (Fig 3-23).



Fig 3-22 "Halo effect" in natural maxillary incisors. Note the similarity in the outline form of the distal surfaces and distoincisal line angles of the maxillary lateral and central incisors.



Fig 3-23 Porcelain veneers on maxillary and mandibular teeth. Note the natural appearance of the maxillary incisal edges with the halo effect. (Porcelain veneers by Gilbert Young, CDT, Plano, Texas.)

Mandibular incisal edge shape

The incisal edge of the mandibular incisor should have a narrow, but defined, flat incisal table. This incisal table should be slightly canted facially. This is referred to as the *pitch* of the incisal table. The facial incisal line angle should be slightly beveled (Becker I, personal communication, 1997) (Fig 3-24; see also Fig 3-21). This incisal edge configuration not only enhances esthetics but also improves function. As the mandible moves forward, the disocclusion occurs efficiently on the leading incisofacial line angle of the mandibular incisor, rather than dragging on the broader facial surface.

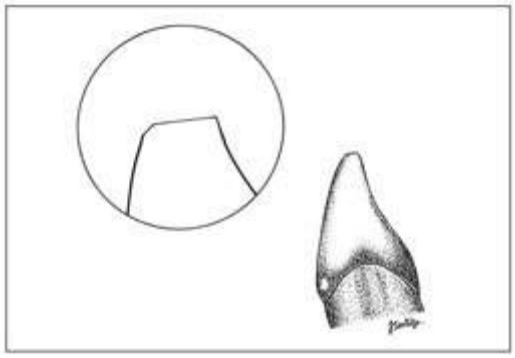


Fig 3-24 The incisal edge of a mandibular incisor. Note the pitch of the incisal table and the bevel of the incisofacial line angle.

Outline symmetry

The distal surfaces of the maxillary central and lateral incisors should be similar in outline form, as should the distoincisal line angles of these teeth[6] (see Fig 3-22). The outline symmetry of the maxillary central and lateral incisors should be similar. A large outline discrepancy (eg, a peg-shaped lateral incisor) negatively affects the beauty of the smile.

Facial contour of the maxillary incisors

The facial surfaces of the maxillary incisors should not be rounded mesiodistally but rather should be flat, with resulting bold mesial and distal line angles and deep facial embrasures (Fig 3-25). Restorations with rounded facial surfaces look unnatural; facial embrasures are not well defined, resulting in a lack of visual distinction of the maxillary anterior teeth.



Fig 3-25 Incisal view of natural maxillary incisors. Note the flat facial surfaces, bold mesial line angles, slightly less bold distal line angles, and deep facial embrasures.

Outline form of maxillary canines

The distal half of the maxillary canine should not be visible when viewed from the front[9] (Fig 3-26). As the eye moves laterally from the midline, each tooth should

appear proportionately narrower than its mesial neighbor. This is termed the *principle of gradation*.[6] After placement of porcelain restorations on the maxillary teeth, the most common offender of this principle is the canine. It appears too wide in relation to both the lateral incisor and the first premolar because the mesiodistal height of contour is too distal.



Fig 3-26 A smile that demonstrates the principle of gradation. The maxillary central incisor is visually the widest tooth, followed by the lateral incisor, the canine, and so on distally. The distal half of the canine must not be visible when viewed from the front to maintain the principle of gradation.

Correct placement of the mesiodistal height of contour on the facial surface of canine restorations involves the skills of both the dentist and the laboratory technician. First, the dentist must remove sufficient tooth structure on the distofacial half of the tooth to allow the technician to create the correct facial contours. Second, the technician must visualize the case from the front during final contouring of the restorations to ensure that the principle of gradation is heeded.

Color

Natural teeth are polychromatic. They generally have higher chroma in the gingival third, transitioning to a lower chroma and higher value in the middle third. The incisal third is characterized by the transition to incisal translucency, which is commonly framed by the halo effect (see Fig 3-22). Maverick colors can appear anywhere, and they individualize the tooth.

The chroma and value of the maxillary lateral incisor is commonly the same as that of the central incisor. In the maxillary canine, the chroma is generally higher, especially in the gingival third, and the value is lower. Incisal translucency is usually minimal in the maxillary canine, and seldom does the halo effect occur.

Polychromaticity in the individual tooth and between neighboring teeth is essential in porcelain restorations if natural beauty is the goal. Color in dentistry is described in greater detail in chapter 4.

Color modifiers

It has been stated that hair color, skin color, and lipstick color all significantly affect shade selection when restorations are being placed in the esthetic zone.[14] Of these modifiers, skin color is by far the most important. A given tooth shade will look lighter and higher in value in a patient with darker skin. Conversely, the same tooth shade will appear yellower and lower in value in a patient with very light skin.

When choosing a tooth shade for a patient with variable skin color, for instance, a white patient with a deep tan, the impact of the skin color must be discussed with the patient prior to treatment. If porcelain restorations are placed while the skin is tanned, the restorations will appear to become more yellow and lower in value as the skin color lightens.

Image

The overall presentation of the smile can be described as the image. Miller[10] discusses the differences between the "natural image" and the "media image." With the media image, the teeth are generally more symmetric, monochromatic, and very high in value. The natural image incorporates asymmetries, polychromaticity, and lower value with higher chroma. Dentists commonly make esthetic choices for patients based on their own notions of beauty rather than on the patient's desires.[18] When restoring maxillary anterior teeth, it is essential that the dentist understand the overall image that the patient desires. To maximize predictability when placing anterior restorations, the issue of overall smile presentation first must be developed to the patient's satisfaction in the provisional restorations.

Age characteristics of teeth

Both tooth color and surface texture relate information about the age of the patient (Figs 3-27 and 3-28).



Fig 3-27 Young teeth demonstrate higher value, lower chro-ma, higher surface texture, and lower luster.



Fig 3-28 Older teeth demonstrate lower value, higher chroma, lower surface texture, and higher luster.

Chroma and value

The value, or brightness, of a tooth is higher in young patients and decreases with age. Conversely, the chroma, or color saturation, is lower in young patients and increases with age.[5]

Surface texture

Surface texture is higher in the young patient and decreases as the patient ages.[5] The surface luster is a function of the amount of surface texture. Therefore, the young tooth with greater surface texture has a lower luster. As the surface texture is worn away with age, the surface luster increases.

It is important to communicate with the laboratory technician about texture and luster. For example, porcelain veneers with low value, low surface texture, and high luster are not appropriate for a 25-year-old patient.

Individual tooth length and proportion

The maxillary central incisors are the centerpiece of the smile.

The average length of the maxillary central incisor is 10 to 11 mm[13] (Fig 3-29). The ratio of height to width in the maxillary central incisor should be approximately 1.2 to 1.0. In other words, the width of the central incisor should be approximately 75% to 80% of its height.[13] When evaluating a smile, the dentist must start with the position and size of the maxillary central incisor. It is difficult, if not impossible, to develop optimum esthetics with short maxillary anterior teeth.



Fig 3-29*(a and b)* Short maxillary central incisors measuring 8 mm due to altered passive eruption. *(c)* After esthetic crown lengthening to treat altered passive eruption.

Tooth-to-tooth proportions

The principle of gradation[6] states that as the eye moves laterally from the midline, each tooth should appear proportionately narrower than its mesial neighbor. There has been much discussion about what this mesiodistal proportion should be. The golden proportion (1.618:1.0), which was formulated as one of Euclid's elements, has been proposed.[19] Viewed from the front, the maxillary central incisor would be 1.618 times wider than the lateral incisor, the lateral incisor would be 1.618 times wider than the visual width of the canine, and so on as the eye moves distally. However, developing esthetic proportions is not that simple. In a patient with a very tapered maxillary arch, the maxillary central incisors will appear wide, and the teeth may approximate the golden proportion. However, in a patient with a very square maxillary arch form, the golden proportion would result in unesthetically wide central incisors. To some degree, the width of the central incisor in relation to that of the lateral incisor is also a matter of personal taste. The golden proportion produces very bold central incisors,[6,20] which appeals to some individuals. However, in natural teeth situated in natural arch forms, the golden proportion seldom occurs. The natural proportion of the width of the maxillary central incisor to that of the lateral incisor, when measured with a caliper, is approximately 1.2 to 1.0.[18] The golden proportion is not based on actual tooth measurements but on the tooth proportions when viewed from the front. This proportion is approximately ¼ to 1.0 in nature (Fig 3-30).

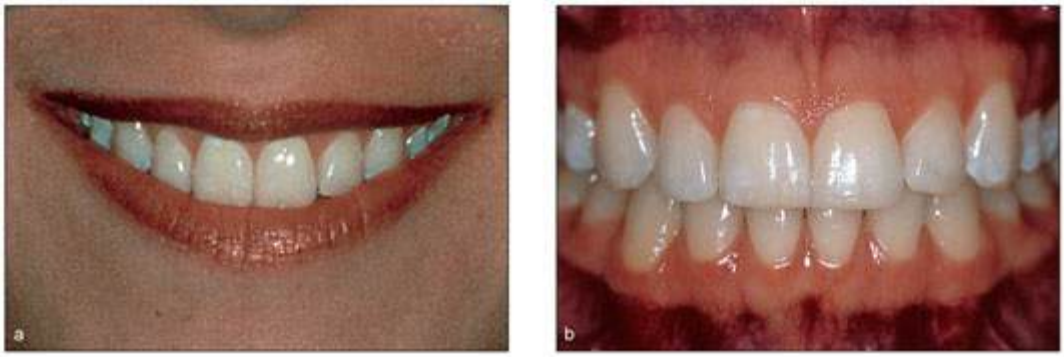


Fig 3-30*(a)* Note the beauty and proportionality of this natural smile. *(b)* The measurement of the visual widths of the maxillary central incisors and maxillary lateral incisosr reveals a natural proportion of central incisor to lateral incisor of ¼ to 1.0.

Because dental esthetics is a matter of taste, the ultimate decision on widths and proportions must be developed in provisional restorations with the patient.

Principle of illumination

Visually, light objects are perceived to approach the viewer and dark objects to recede from the viewer.[6] This principle must be considered when high-value porcelain or resin composite restorations are placed only on maxillary anterior teeth, because the result may be an unesthetic visual separation of the anterior and posterior teeth. A visual coupling of the front and back of the mouth may require placement of restorations on one or more maxillary premolars (Fig 3-31).

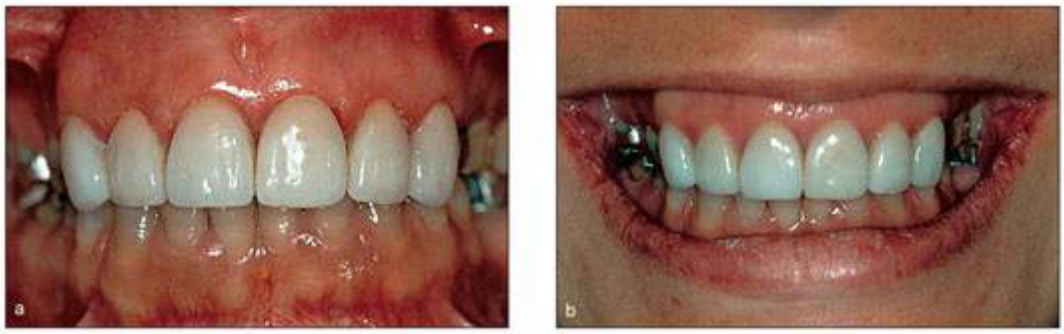


Fig 3-31*(a and b)* Porcelain veneers were placed only on the maxillary anterior teeth. Note that the maxillary first premolars are virtually invisible when viewed from the front. This results in a loss of visual coupling of the front and the back of the mouth. Note also the lower lip asymmetry.

Law of the face

The face of a tooth is that portion of the facial surface bound by transitional line angles when viewed from the front.[5] To make teeth of dissimilar widths appear similar, the apparent faces should be made equal.

To make an anterior tooth appear wider, the transitional facial line angles are moved into the interproximal facial embrasures. Conversely, to make an anterior tooth appear narrower, the transitional line angles are moved closer to the tooth midline (Fig 3-32).

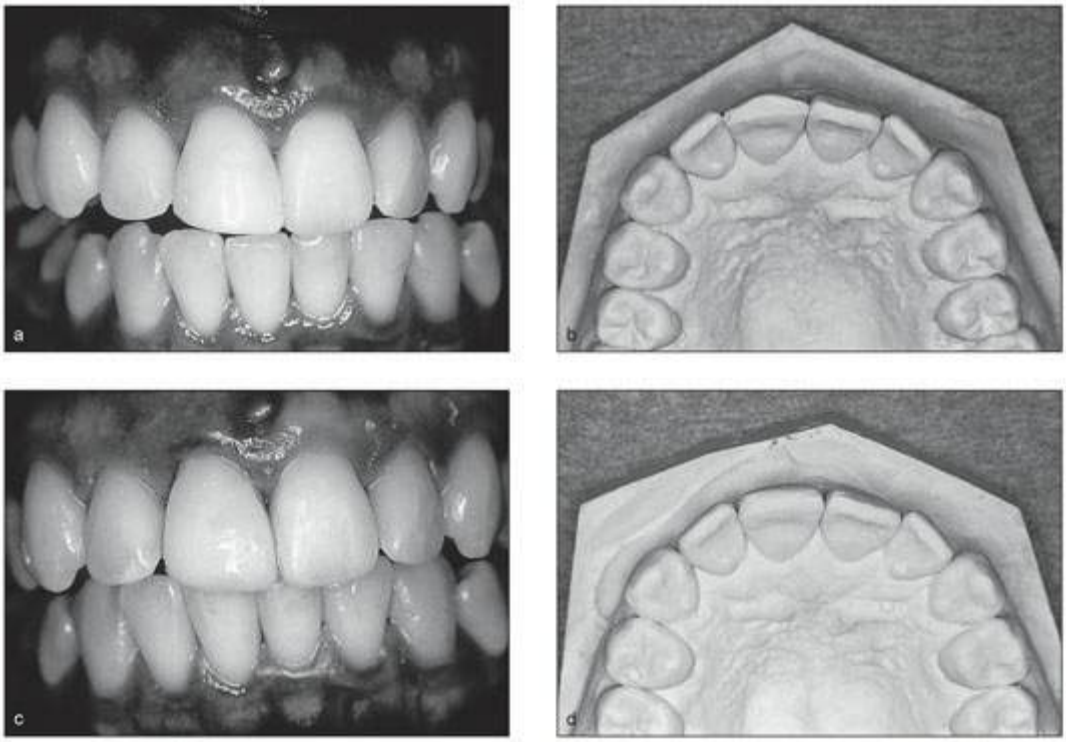


Fig 3-32*(a)* A direct bonded restoration on the maxillary right central incisor appears too wide. *(b)* The cast of the maxillary arch demonstrates that the mesiofacial and distofacial transitional line angles are too far into the facial embrasures, resulting in a visual widening of the tooth. *(c)* After replacement of the direct bonded restoration on the right central incisor, the tooth now appears narrower. *(d)* The cast of the patient after the direct bonded restoration was replaced. Note the narrower appearance of the tooth because the mesiofacial and distofacial transitional line angles are closer together.

Conclusion

The provision of esthetic and functional restorative dentistry must be based on a set of guidelines founded on the best clinical science available. These guidelines can then be used to determine a patient's overall orofacial presentation and to guide the diagnostic wax-up. Based on the diagnostic wax-up, the provisional restorations are fabricated and placed. It is at this point that the "eye of the artist" becomes helpful for developing the final shade, shape, contour, and incisal edge configuration of the provisional restorations, which will serve as the blueprint for the defintive restorations.

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